Foundations of Physical AI

How AI learns to sense, reason, and act in the physical world

30-minute foundation + 20-minute case study module

From digital prediction to physical action

Physical AI is the next step after generative AI: outputs become movements, controls, and interventions.

Digital AI

predicts, writes, classifies, reasons over files

→

Physical AI

perceives, plans, controls, learns from feedback

→

Action

robots, vehicles, smart spaces, machines

Key idea: the output is no longer only text or pixels. It becomes a safe physical behavior under real constraints.

SLIDE 03

Thirty-minute roadmap

A compact foundation: what it is, how it works, how to deploy it safely.

1

Define Physical AI

What makes it different from digital AI

2

Core building blocks

Sensing, world models, policies, control

3

Training workflow

Real data, simulation, synthetic data, evaluation

4

Safety and deployment

Risk controls, metrics, governance, roadmap

Suggested pacing

4m

10m

8m

8m

SLIDE 04

Working definition

Physical AI systems perceive the physical world, reason about it, and perform actions through machines.

Sensors
vision
force
position

→

Models
world state
intent
uncertainty

→

Policies
plans
actions
control

→

Actuators
wheels
arms
grippers

Closed-loop autonomy

The system constantly compares intended behavior with what actually happened, then updates its next action.

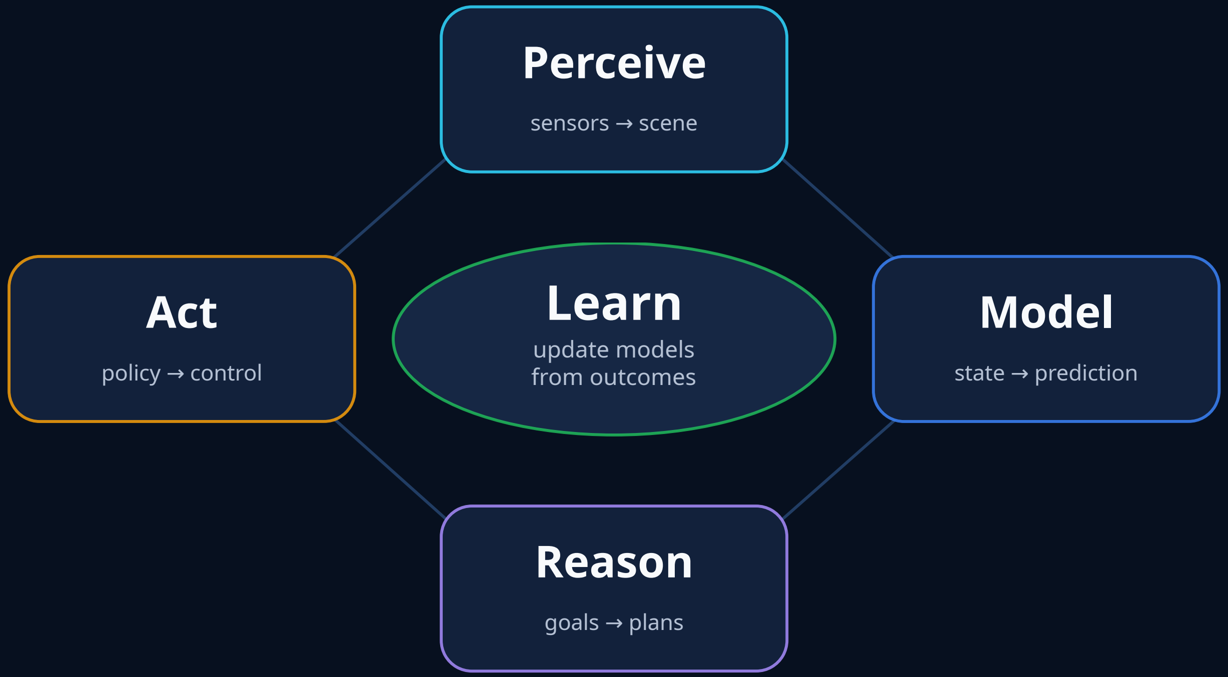
Concept aligned with NVIDIA's definition of Physical AI: autonomous systems that perceive, understand, reason, and act in the physical world.

The core loop

Every Physical AI system is built around a feedback loop between perception and action.

“Intelligence” emerges from the loop, not from a model alone.

Failure at any layer can break the system: bad sensing, wrong state, unsafe plan, weak control, or poor feedback.



Why the physical world is hard

The same model must handle uncertainty, contact, latency, safety, and long-tail edge cases.

Partial observability	sensors miss, occlude, drift
Physics and contact	friction, force, deformable objects
Real-time constraints	milliseconds matter for safety
Irreversible errors	a bad action can damage people or equipment
Long-tail environments	rare events dominate failure modes

A robot policy is useful only when it is robust under noisy sensing and imperfect execution.

The Physical AI stack

Think of it as a layered system, from body and sensors up to policies and supervision.

Governance & safety case	risk controls, audits, standards, incident response
Task planning & policy	goals → actions; skill selection; language grounding
World model / state estimation	what is where, what may happen next, uncertainty
Perception	camera, lidar, force, audio, proprioception
Control & actuation	motors, arms, grippers, wheels, low-level controllers
Embodiment & environment	morphology, workspace, objects, people, constraints

Good systems are rarely “one model.” They are engineered stacks that combine learned components with control, monitoring, and safety logic.

Grounding: connecting words to the world

Physical AI must translate human goals into geometric, semantic, and actionable representations.

Instruction

"Pick up the red part and place it in bin A."

Scene

objects, positions, people, obstacles

Affordance

what can be grasped, moved, opened

Action

trajectory + force + timing

Geometry

3D maps, poses, distance

Semantics

object identity, task meaning

Dynamics

motion, force, cause-effect

Uncertainty

confidence, hidden state, risk

Foundation models add generalization

The trend is toward models that transfer knowledge across tasks, objects, and robot bodies.

Vision-language models

interpret images, objects, scenes, and instructions

Vision-language-action models

map observations and commands toward robot actions

Generalist robot policies

pretrained across many tasks and embodiments, then fine-tuned

Research examples

- **RT-2** — web-scale vision-language knowledge transferred into robotic control
- **Open X-Embodiment** — 22 robots, 527 skills, shared datasets for cross-robot transfer
- **Octo** — open generalist manipulation policy trained on 800k robot episodes

Caveat: bigger models do not remove the need for control, safety limits, and task-specific validation.

Sources: Google DeepMind RT-2; Open X-Embodiment; Octo project pages.

World models and simulation

A system needs a place to practice safely before it touches production equipment.

> Simulation reduces cost and risk, but production validation remains required because no simulator is complete.

Digital twin
geometry, materials, constraints

Physics simulation
motion, contact, sensors

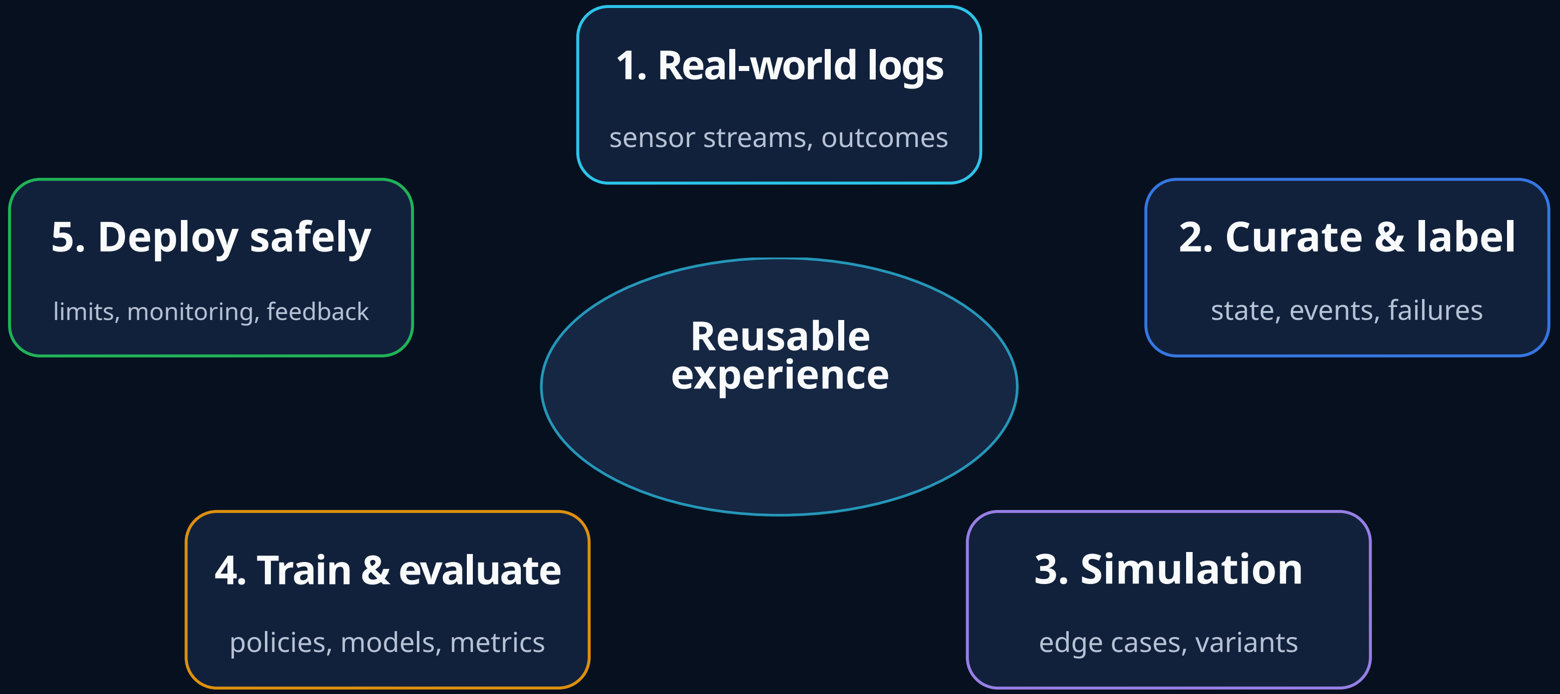
World model
predict likely futures

Policy training
learn actions safely

Source: NVIDIA Cosmos describes world foundation models for generation, reasoning, and synthetic data in Physical AI workflows.

The Physical AI data flywheel

Progress depends on converting physical experience into reusable training and evaluation data.



The strongest teams design the data loop before they scale the robot fleet.

How Physical AI systems learn

No single learning method is sufficient; practical systems combine several.

Imitation learning	learn from human demonstrations or teleoperation	fast start; limited by demonstration coverage
Reinforcement learning	learn by optimizing reward in simulation or constrained real trials	can discover skills; reward design is hard
Self-supervised learning	predict future frames, state changes, contacts, or affordances	uses raw data; needs robust evaluation
Fine-tuning / adaptation	adjust a pretrained policy to a site, robot, or task	practical but can overfit or forget

Practical recipe **pretrain broadly → demonstrate task → train in simulation → validate on hardware → monitor drift**

Crossing the sim-to-real gap

The policy trained in simulation must survive the imperfections of the real machine and environment.

Simulation

controlled physics
perfect labels
cheap retries
rare-event generation

Transfer toolkit

randomize, calibrate, constrain, test

Real world

wear and tear
lighting changes
sensor noise
human behavior

- **Domain randomization**
- **System identification**
- **Safety envelopes**
- **Hardware-in-loop tests**

Architecture choices

Physical AI can be built as modular pipelines, end-to-end policies, or hybrid systems.

Modular

perception → planning → control

interpretable; easier safety review
brittle interfaces between modules

End-to-end

observation → action

less hand engineering
harder to verify and debug

Hybrid

foundation model + controllers + safety layer

practical balance
still needs integration discipline

Most industrial deployments today favor hybrid architectures: learned perception and planning with explicit safety and control layers.

Evaluation is multidimensional

A useful robot is not just accurate. It must be safe, robust, fast enough, and maintainable.

Task success

completion rate, time, quality

Generalization

new objects, layouts, lighting, users

Robustness

sensor failures, disturbances, rare events

Latency

perception-to-action delay

Safety

near-misses, force limits, stop behavior

Human factors

handover, trust, supervision workload

> Use “release gates”: a policy advances only when it passes offline, simulated, hardware-in-loop, supervised, and production metrics.

Why now: robotics is scaling while AI capability rises

Industrial adoption creates a practical base for data, deployment, and automation economics.

542k

industrial robots installed in 2024

IFR, 2025

4.664M

industrial robots operating worldwide

IFR, 2025

54%

of 2024 global deployments in China

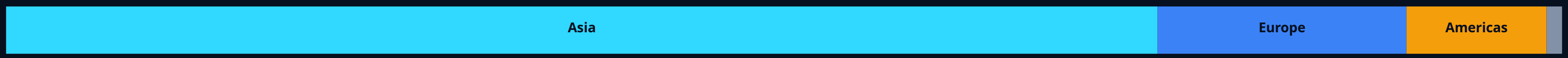
IFR, 2025

74%

of new deployments in Asia

IFR, 2025

2024 regional deployment split



The opportunity is not just "more robots." It is reusable robot intelligence across more machines and sites.

Source: International Federation of Robotics, World Robotics 2025 report. See references.

Safety and governance are part of the foundation

Physical AI needs a safety case, not just a model card.

Govern accountability, roles, human oversight

Map hazards, operating domain, affected stakeholders

Measure tests, simulations, stress cases, incident metrics

Manage controls, monitoring, escalation, shutdown

NIST AI RMF voluntary risk framework for trustworthy AI

ISO 10218-1:2025 industrial robot safety requirements

EU AI Act risk-based obligations for selected AI systems

Minimum practice: define the operating design domain, hard safety limits, test coverage, human override, logging, and post-deployment monitoring.

Sources: NIST AI RMF; ISO 10218-1:2025; European Commission AI Act.

Where Physical AI shows up first

The early value is strongest where perception, motion, and operational feedback are already instrumented.

Manufacturing	inspection, assembly, machine tending
Warehouses	AMRs, picking, packing, inventory
Autonomous vehicles	perception, planning, scenario testing
Healthcare robotics	assistive, surgical, lab automation
Smart spaces	traffic flow, safety, monitoring

Pattern: start with bounded environments, measurable tasks, and clear safety constraints.

A practical starting roadmap

Treat Physical AI as an engineered product lifecycle, not a lab demo.

1. **Bound the task** — choose one workflow and operating domain
2. **Instrument the environment** — collect sensor, outcome, and failure data
3. **Build a digital testbed** — simulate normal and rare scenarios
4. **Train and gate releases** — offline → sim → hardware → supervised deployment
5. **Operate the flywheel** — monitor, retrain, audit, and improve

> Do not begin with “build a humanoid.” Begin with a narrow task where failure modes can be named, tested, and controlled.

Current limitations and open problems

The field is advancing quickly, but several foundations remain difficult.

Data scarcity

robot data is expensive, heterogeneous, and hard to share

Brittle generalization

policies can fail under small environmental changes

Verification gap

hard to prove learned behavior is safe in all cases

Compute and latency

large models may not fit real-time edge constraints

Human trust

users need predictable behavior and clear override paths

Governance

updates can change behavior after initial validation

Near-term progress will come from narrower operating domains, stronger data flywheels, better simulation, and explicit safety architecture.

Three takeaways

Use these as the closing frame for the discussion.

1. **Physical AI is closed-loop autonomy**
   It connects perception, state estimation, planning, control, and learning.

2. **Foundation models help, but do not replace engineering**
   The model must sit inside a tested stack with controllers and safety limits.

3. **Deployment is a data-and-safety flywheel**
   Instrumentation, simulation, evaluation, monitoring, and governance drive improvement.

Discussion prompt: What task in your environment is bounded enough to test, but valuable enough to matter?

20-MINUTE CASE STUDY

Amazon Robotics as Physical AI

How a large logistics network combines robots, AI routing, tactile manipulation, human supervision, and operational data.

1M+	**+10%**	**25%**
robots deployed	robot travel efficiency target	processing-time reduction

Sources: Amazon Robotics/Operations updates on DeepFleet, Vulcan, and next-generation fulfillment centers.

Why this case is useful

Amazon's fulfillment network is a bounded but highly variable physical environment.

1. **Scale** — Large robot fleet creates enough operational data to improve routing and system behavior.
2. **Embodiment** — Mobile robots, arms, bins, conveyors, shelves, workers, and packages form one physical system.
3. **Feedback** — Every movement can produce logs: location, delay, exception, handoff, and outcome.
4. **Safety boundary** — The environment is designed with zones, workstations, procedures, and human overrides.

The lesson is not "copy Amazon." The lesson is how Physical AI becomes deployable when the task, environment, metrics, and escalation paths are engineered together.

System map: from order to physical action

The intelligence is distributed across planning software, robot policies, controls, and human operators.

> Physical AI case studies should be read as systems, not as single-model demos.

Demand
orders, promises, inventory signals

Tasking
which item, bin, robot, station

Movement
routes, traffic, charging, congestion

Manipulation
pick, stow, sort, pack

Human handoff
exceptions, coaching, supervision

Operational data flywheel
logs → exceptions → retraining → deployment gates → monitoring

Fleet-level Physical AI: DeepFleet

At fleet scale, routing becomes a learning problem: reduce congestion, travel time, and idle motion.

1M+
robots in operations

+10%
reported robot travel efficiency

300+
facilities in robot network

Core idea
traffic management for robots

Observe
inventory movement and robot state

→

Predict
bottlenecks and travel delays

→

Optimize
routes and fleet coordination

→

Improve
logs feed model updates

> Do not overread the metric: Amazon reports robot travel efficiency improvement, not a universal 10% improvement in all fulfillment outcomes.

Source: Amazon, “AI foundation model to power its robotic fleet,” June 30, 2025.

Tactile Physical AI: Vulcan

Manipulation becomes harder when contact is expected, not avoided.

Vision locate the item and the grasp region

Touch measure contact force and torque

Control move, slide, back off, or replan

Handoff ask a human when the item is unsuitable

Case insight: touch changes the policy. The robot can use contact as information, instead of treating contact only as a fault.

Sources: Amazon Vulcan launch; Amazon Science technical explainer on six-axis force/torque sensing.

Workflow integration: AI plus robotics plus layout

The site design matters as much as the robot model.

3M+
square feet in Shreveport site

30M+
items supported by Sequoia

25%
reported processing-time reduction

25%
targeted cost-to-serve improvement

Store → Bring pods → Pick/stow → Sort → Pack → Ship

Physical AI deployment often succeeds through co-design: robot capability, building layout, workstation ergonomics, telemetry, and exception handling.

Source: Amazon, next-generation fulfillment center update, October 2024.

Human-in-the-loop safety model

Physical AI at scale still depends on designed handoffs, training, and clear operating boundaries.

Operational controls

Operating domain	bounded warehouse areas, stations, pods, and procedures
Exception handoff	robot asks for human help when confidence or feasibility is low
Ergonomics	robots bring work into safer reach zones where possible
Monitoring	logs expose delays, near-misses, failures, and maintenance needs

Workforce design

700k+

employees upskilled

30%

more technical support roles in advanced sites

Use this critically: workforce benefits depend on implementation, job design, transparency, and local labor conditions.

> Safety is not only collision avoidance. It includes ergonomic risk, exception handling, training, maintenance, and governance of software updates.

Sources: Amazon DeepFleet launch; Amazon next-generation fulfillment center update.

Reported outcomes — and what to question

A good case study includes both evidence and limits.

Reported by Amazon

Scale	more than 1 million robots deployed
Fleet efficiency	DeepFleet targets 10% better robot travel efficiency
Site throughput	up to 25% lower fulfillment processing time in next-gen sites
Human workflow	ergonomic workstations and more technical support roles

Critical reading

Metric scope	robot travel efficiency is not total order-cycle efficiency
Evidence	most published figures are company-reported
Transferability	warehouse logistics is more bounded than homes, hospitals, or roads
Governance	software updates can change behavior after validation

Case-study discipline: separate operational claims, independent evidence, and design lessons.

Sources: Amazon Robotics/Operations updates; interpretation is based on scope of the reported metrics.

Transferable design lessons

What another organization can borrow from the case without copying Amazon’s scale.

1	**Bound the operating domain**	Define where the robot may operate, what objects it can handle, and when it must stop.
2	**Instrument before scaling**	Capture task state, robot state, failure modes, and human interventions from day one.
3	**Use a hybrid stack**	Combine learned models with controls, safety monitors, routing rules, and human escalation.
4	**Co-design the environment**	Layout, bins, workstations, lighting, and labels can reduce the AI burden.
5	**Treat data as the product**	The durable asset is not only the robot; it is the data flywheel that improves behavior.

Case study readout

Physical AI becomes useful when intelligence is constrained, measured, and connected to operations.

1. **Fleet intelligence matters** — The routing layer can create value even when individual robots are simple.
2. **Touch extends autonomy** — Contact-rich manipulation needs force sensing, control, and safe fallback.
3. **The system is the product** — Robots, humans, layout, data, monitoring, and governance form the deployable unit.

Discussion prompt: Where could your organization build a bounded Physical AI pilot with a measurable data flywheel?

References and further reading

Key sources used for the deck.

1. NVIDIA Glossary — What is Physical AI? https://www.nvidia.com/en-us/glossary/generative-physical-ai/

2. Google DeepMind — RT-2: New model translates vision and language into action. https://deepmind.google/blog/rt-2-new-model-translates-vision-and-language-into-action/

3. Open X-Embodiment — Robotic Learning Datasets and RT-X Models. https://robotics-transformer-x.github.io/

4. Octo — An Open-Source Generalist Robot Policy. https://octo-models.github.io/

5. NVIDIA Cosmos — Physical AI with World Foundation Models. https://www.nvidia.com/en-us/ai/cosmos/

6. International Federation of Robotics — World Robotics 2025 report. https://ifr.org/ifr-press-releases/news/global-robot-demand-in-factories-doubles-over-10-years

7. Stanford HAI — 2025 AI Index Report. https://hai.stanford.edu/ai-index/2025-ai-index-report

8. NIST — AI Risk Management Framework. https://www.nist.gov/itl/ai-risk-management-framework

9. ISO — ISO 10218-1:2025 Robotics safety requirements. https://www.iso.org/standard/73933.html

10. European Commission — AI Act. https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai

11. Amazon — Amazon launches a new AI foundation model to power its robotic fleet and deploys its 1 millionth robot. https://www.aboutamazon.com/news/operations/amazon-million-robots-ai-foundation-model

12. Amazon — Amazon has more than 1 million robots that sort, lift, and carry packages. https://www.aboutamazon.com/news/operations/amazon-robotics-robots-fulfillment-center

13. Amazon — Introducing Vulcan: Amazon's first robot with a sense of touch. https://www.aboutamazon.com/news/operations/amazon-vulcan-robot-pick-stow-touch

14. Amazon Science — How Amazon's Vulcan robots use touch to plan and execute motions. https://www.amazon.science/blog/how-amazons-vulcan-robots-use-touch-to-plan-and-execute-motions

15. Amazon — Next-generation fulfillment center powered by AI and robotics. https://www.aboutamazon.com/news/operations/amazon-fulfillment-center-robotics-ai

16. Amazon IR — Q2 2025 results, DeepFleet update. https://ir.aboutamazon.com/news-release/news-release-details/2025/Amazon-com-Announces-Second-Quarter-Results/

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